

Lecture 8

- Electric (Scalar) Potential

A force field is said to be **conservative** if the work done to move an object between two points depends only on the starting and ending positions, not on the specific path taken.

Equivalently, the net work done by the field around any closed loop is zero.

Electric Potential

Electrostatic field

$$\textcircled{1} \quad \nabla \cdot \vec{D} = \rho_v$$

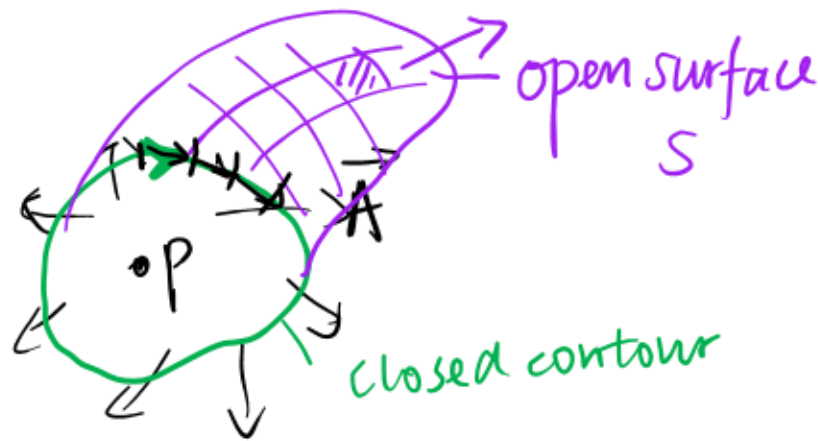
$$\textcircled{2} \quad \nabla \times \vec{E} = 0$$

$$\vec{D} = \epsilon \vec{E}$$

$\vec{E}$ is conservative.

differential form

Stokes' theorem



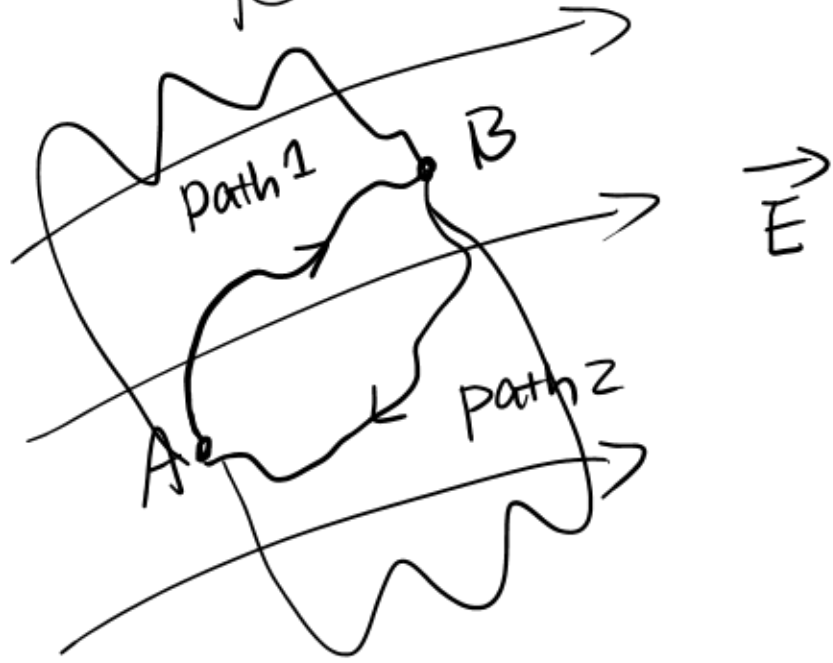
$$\oint_C \vec{A} \cdot d\vec{C} = \int_{\text{open surface}} (\nabla \times \vec{A}) \cdot d\vec{S}$$

Integral form

$$\oint_C \vec{E} \cdot d\vec{C} = \int_{\text{open surface}} (\underbrace{\nabla \times \vec{E}}_0) \cdot d\vec{S} = 0$$

Electric Potential

$$\oint_C \vec{E} \cdot d\vec{l} = 0$$



$$\Rightarrow \oint \vec{E} \cdot d\vec{l}$$

$$= \int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l}$$

path 1 path 2

$$= \int_A^B \vec{E} \cdot d\vec{l} - \int_A^B \vec{E} \cdot d\vec{l} = 0$$

path 1 path 2

$$\int_A^B \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l}$$

path 1 path 2

$\Rightarrow$ the line integral of $\vec{E}$ between two points is path independent.

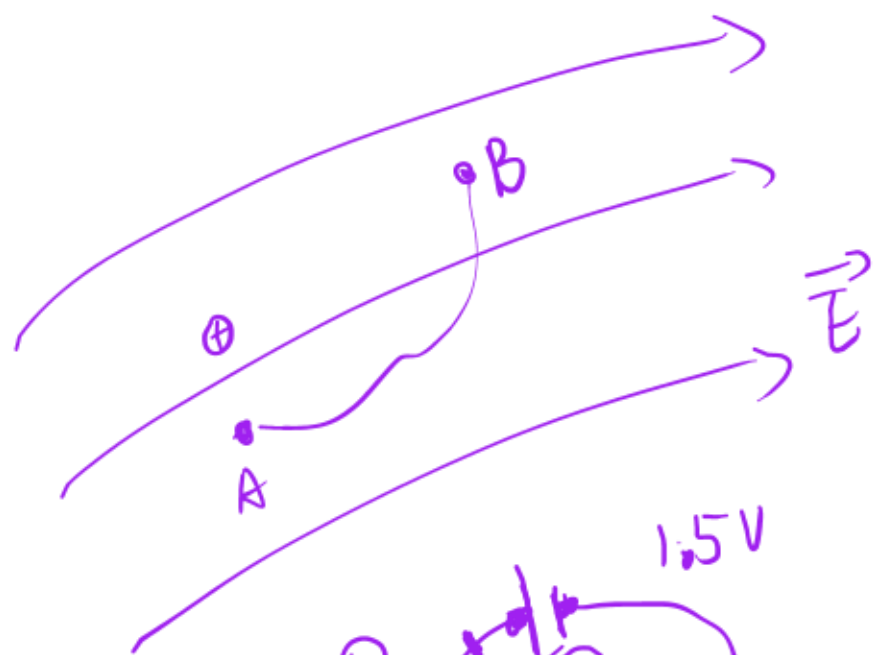
Electric Potential

[voltage]

$$\textcircled{2} [V] = \left[\frac{N}{C} \cdot m \right]$$

Electric Potential Difference from A to B

$$\Rightarrow \left[\frac{N}{C} \right] = \left[\frac{V}{m} \right]$$



$$\underline{V_B - V_A} = \textcircled{-} \int_A^B \vec{E} \cdot d\vec{l}$$

$$= - \frac{1}{q} \int_A^B q \vec{E} \cdot d\vec{l}$$

$$V_B - V_A < 0$$

$$V_B < V_A$$

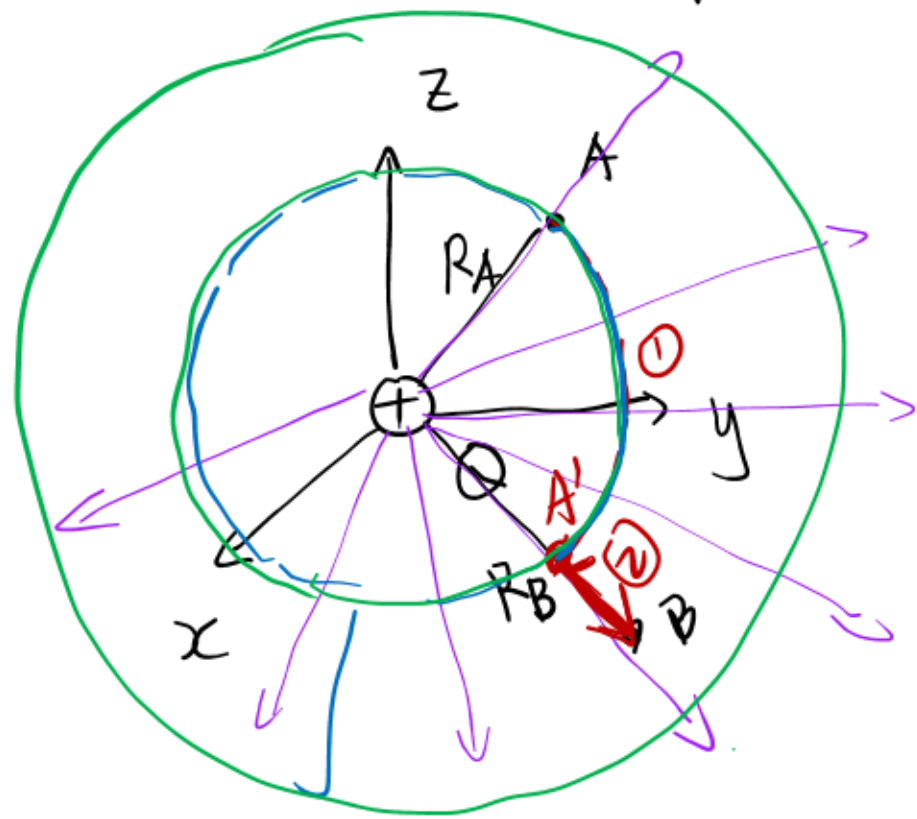
$$\oint \vec{E} \cdot d\vec{l} = 0$$

Note -



Electric Potential

Example



Equipotential surface.

$$V_B - V_A = - \int_A^B \vec{E} \cdot d\vec{l}$$

$$\vec{E} = \frac{Q}{4\pi\epsilon R^2} \hat{R}$$

flexibility of choose path
path segment $\parallel \vec{E}$
path seg $\perp \vec{E}$

$$= \int_{A'}^{A'} \vec{E} \cdot d\vec{l} - \int_{A'}^B \vec{E} \cdot d\vec{l}$$

$$= - \int_{A'}^B \frac{Q}{4\pi\epsilon R^2} \cdot \hat{R} \cdot \hat{R} dR$$

$$= - \int_{R_A}^{R_B} \frac{Q}{4\pi\epsilon R^2} dR$$

$$= \frac{Q}{4\pi\epsilon} \left. \frac{1}{R} \right|_{R_A}^{R_B}$$

$$V_B - V_A = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_B} - \frac{1}{R_A} \right)$$

Define ∞ as a reference point
 $V = \frac{Q}{4\pi\epsilon R}$

$$\hat{R} \cdot (\hat{\theta} \dots + \hat{\phi} \dots) = 0$$

$$\int -\frac{1}{R^2} dR = \frac{1}{R}$$

Electric Potential

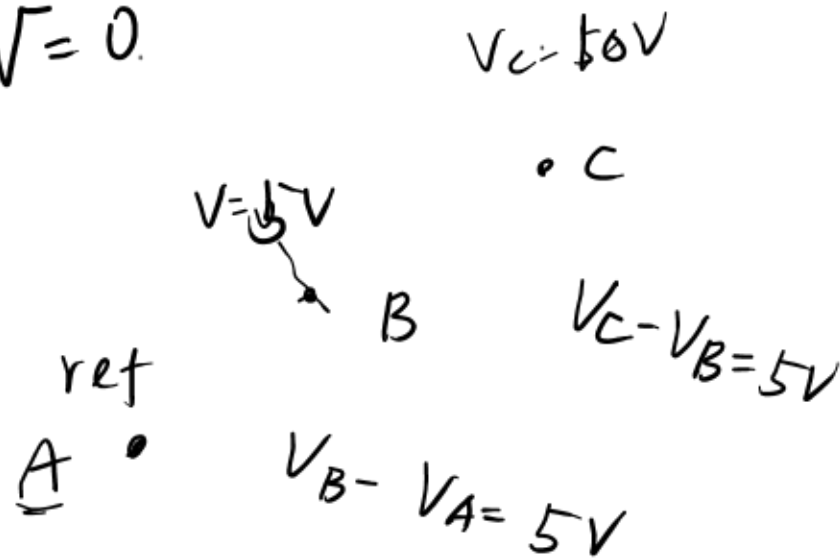
Electric potential at a point is the potential difference with respect to a reference point that we assign $V=0$ (ground)

Potential at a point $P =$

$$V_P - V_{\text{ref}} = - \int_{\text{ref}}^P \vec{E} \cdot d\vec{l}$$

$\Downarrow$
 $V=0$

$$V_P = - \int_{\text{ref}}^P \vec{E} \cdot d\vec{l}$$



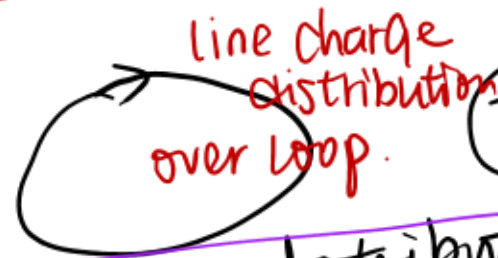
Electric Potential

- ① Potential for point charge at origin with reference point at ∞

$$V = \frac{Q}{4\pi\epsilon R}$$

- ② Electric field will always be normal to equipotential surface.

- ③ For finite charge distribution, we can take ∞ as a ref point



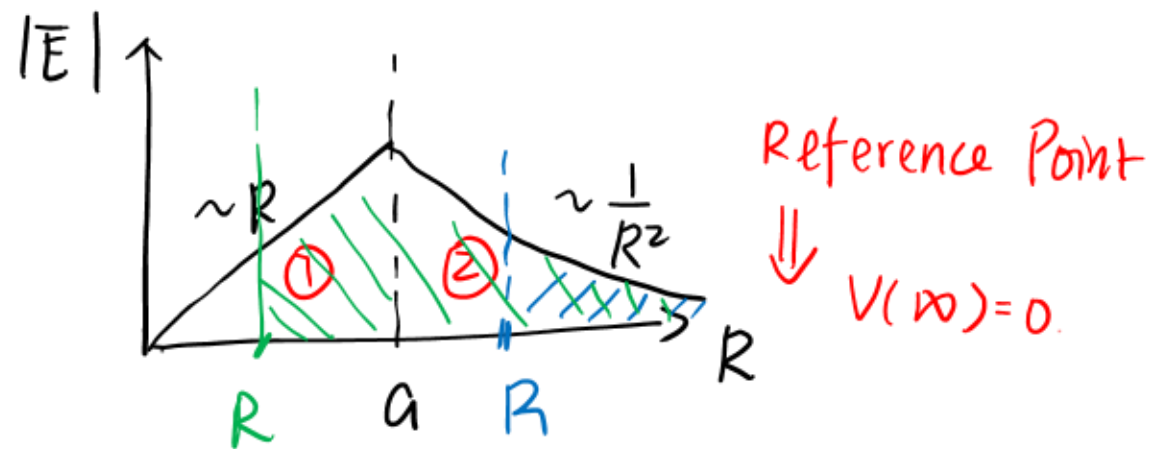
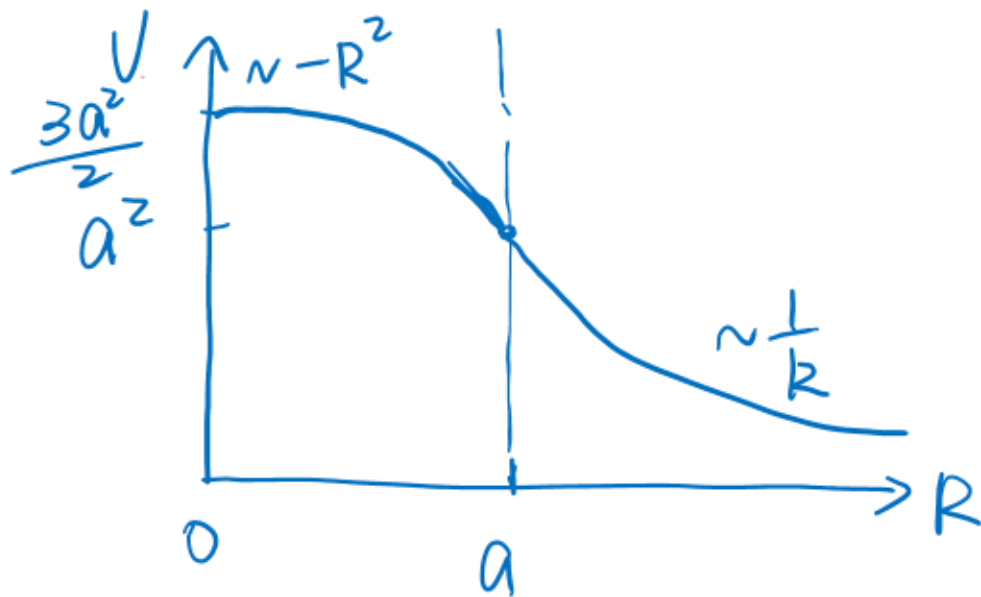
But for infinite charge distribution we cannot take ∞
Example next class.

Electric Potential

Example = Given electric field

$$\vec{E} = \begin{cases} \hat{R} R, & R \leq a \\ \hat{R} \frac{a^3}{R^2}, & R > a \end{cases}$$

Calculate V given $V(\infty) = 0$



Always, start from reference point $V = 0$

$$\begin{aligned} \text{Region ②: } V(R) - \underbrace{V(\infty)}_{\text{ref.}} &= - \int_{\infty}^R \hat{R} \frac{a^3}{R^2} \cdot \hat{R} dR \\ &= - \int_{\infty}^R \frac{a^3}{R^2} dR = \frac{a^3}{R} \quad (R > a) \end{aligned}$$

$$\begin{aligned} \text{Region ①: } V(R) - \underbrace{V(\infty)}_{\text{ref.}} &= - \int_{\infty}^a \vec{E} \cdot d\vec{l} - \int_a^R \vec{E} \cdot d\vec{l} \\ &= a^2 - \int_a^R R dR = -\frac{R^2}{2} + \frac{3a^2}{2} \quad (R \leq a) \end{aligned}$$